

Multi-Service Computational Thread

A Pareto implementation plan for evidence-backed agent development across local repositories

IMPLEMENTATION IN PROGRESS	25 AUGUST 2026	S0-S3 INDEPENDENTLY VERIFIED: PASS
----------------------------	----------------	------------------------------------

The decision

Build a read-only Computational Thread above existing sessions. The OpenAPI fit gate correctly stopped; the selected Kotlin descriptor contour now composes exact projected shapes, declared cross-repository impact, and conditional change-set coverage.

Why this is the 80/20 choice

It targets the failure mode where ordinary grep and LLM navigation are weakest: contract names differ across providers and consumers, revisions are easy to mix, and omitted repositories are difficult to detect. It reuses Codeclew's existing session, CAS, certainty, and bounded-context authorities.

Hard boundary

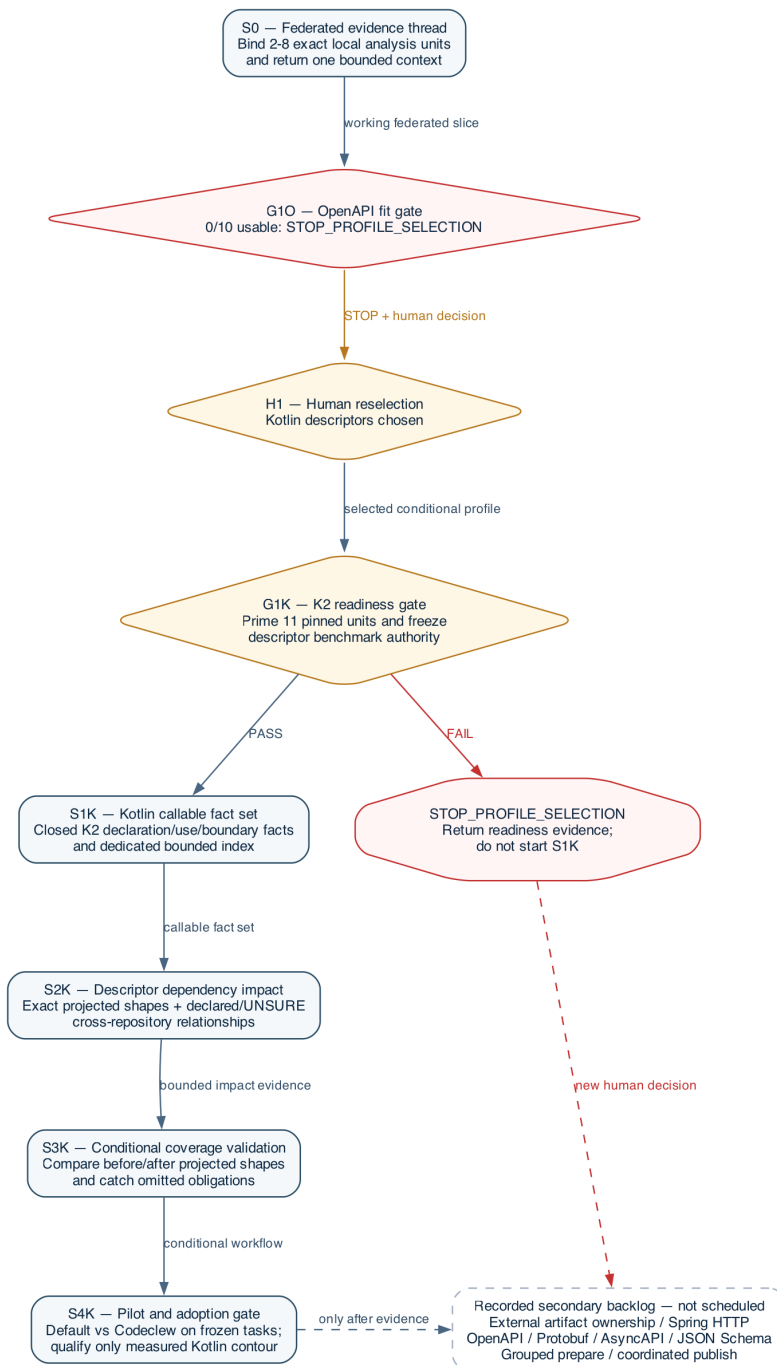
The core plan is local, explicit-membership, and read-only. It does not execute multi-repository candidates, coordinate publication, infer runtime or HTTP behavior, or add framework-specific meaning without separate evidence.

Current checkpoint

S0 through S3 are implemented and independently verified. S4 must first freeze an unambiguous measurement protocol, then run the one-shot Default-versus-Codeclew pilot. No adoption claim is allowed before that gate passes.

Roadmap at a glance

The main path has six bounded steps and one early stop decision. The dashed branch records possible future work; it is not part of the implementation commitment.



Canonical source: docs/plans/codeclew-multiservice-thread-plan.dot. A G1 failure returns to a human profile decision; it does not silently expand scope.

Detailed verified plan

The following sections preserve the complete implementation decision record, including scope, authority, limits, per-step necessity, verification, Definition of Done, Pareto deferrals, and approval gates.

Codeclew Multi-Service Computational Thread — Kotlin Descriptor Navigation Plan

Document status

- **Status:** S0, G1O, H1, G1K, S1K, S2K, and S3K completed. S4K protocol closure is next; measured pilot arms have not started.
- **Date:** 2026-08-25.
- **Execution order:** S0 -> G1O(STOP) -> H1 -> G1K -> S1K -> S2K -> S3K -> S4K.
- **Primary outcome:** An agent can bind exact evidence from 2-8 local analysis units, inspect one bounded cross-repository context, navigate compiler-proven Kotlin callable shapes and uses, compare exact projected shapes without overclaiming service-contract meaning, and validate that a proposed multi-repository investigation acknowledges both selected pair members and every unresolved obligation.
- **Product boundary:** The workflow is read-only. Cross-repository topology is DECLARED_TOPOLOGY unless a future, separately approved authority proves artifact ownership. Kotlin descriptor exactness never implies HTTP routing, behavioral compatibility, source compatibility, or binary compatibility.

Decision history

The original Pareto plan selected OpenAPI 3.x as its first exact contract profile, subject to a pre-implementation artifact-fit gate. That gate, now named G1O, froze ten tasks over eight local service pairs and found no tracked OpenAPI provider or consumer authority: 0 EXACT_COMPARABLE, 0 DECLARED_NAVIGABLE, and 10 NOT_USABLE_FOR_V1. Its correct result was STOP_PROFILE_SELECTION.

At H1 the user selected **Kotlin descriptors**. This is a replacement product contour, not a relabelling of the failed OpenAPI evidence. Independent review approved a deliberately conditional Kotlin navigation profile provided that:

1. exactness is limited to a closed K2 compiler-projected declaration shape;
2. cross-repository relations remain declared or unsure;
3. Spring/HTTP interpretation remains excluded;
4. a new G1K gate proves every pinned unit can produce sealed K2 facts and freezes a descriptor-specific benchmark before S1K implementation.

Product decisions

D1 - Status: implemented Why: Not selected:
D2 - Status: implemented Why: Not selected:

D3 - Status: implemented/required Why: Not selected:
D4 - Status: implemented Why: Not selected:
D5 - Status: required Why: Not selected:
D6 - Status: required Why: Not selected:
D7 - Status: required Why: Not selected:
D8 - Status: implemented Why: Not selected:
D9 - Status: required Why: Not selected:
D10 - Status: required Why: Not selected:

Scope

Included

- 2-8 explicitly selected local analysis units, including mixed-language S0 composition and Kotlin-only descriptor qualification.
- Immutable thread and context authority with exact session, revision, generation, query-index, and evidence references.
- Compiler-backed Kotlin FUNCTION, CONSTRUCTOR, CLASS, PROPERTY, and MUTABLE_PROPERTY descriptors plus supported K2 relation/use facts.

- Exact lookup by full symbol identity, navigation lookup by CallableId family and bounded identifier terms, and explicit overload ambiguity.
- A thread-owned, content-addressed callable fact set and query index.
- Read-only before/after projected-shape observations and conditional change-set coverage validation.
- A frozen Default-versus-Codeclew value pilot over the same ten task IDs.

Excluded

- OpenAPI, Protobuf, AsyncAPI, GraphQL, JSON Schema, database/configuration contracts, and remote reference resolution.
- Spring annotations, composed annotations, route paths/methods, HTTP status or payload semantics, Feign/generated-client ownership, Jackson/serialization, validation, security, DI, and framework startup.
- Kotlin source/binary compatibility verdicts; parameter names/defaults; inline/reified/suspend behavior; value-class boxing; reflection; type-alias semantics; Java declaration authority; generated/local declarations.
- Automatic topology discovery, dynamic plugins, daemon, graph database, UI, watcher, candidate creation, task execution, mutation, publication, or rollback across repositories.

Global invariants

- **I1 — Session isolation:** Existing session schemas and one-repository lifecycle remain authoritative and unchanged in meaning.
- **I2 — Exact inputs:** Every member, fact, comparison, and verdict binds exact session, revision, snapshot, generation, compilation, runtime, extractor, profile, and evidence digests.
- **I3 — No certainty upgrade:** Partial, boundary, ambiguous, lexical, or truncated evidence never becomes exact through aggregation.
- **I4 — Global bounds:** All thread/profile/fact/query/source/stdout budgets are thread-global, explicit, and fail closed.
- **I5 — Read-only:** No thread command creates a candidate, starts a task run, updates a ref, or publishes a repository.
- **I6 — No new target execution:** G1K may prime existing qualified member generations. Warm S1K-S3K starts no Gradle, Maven, Java, Kotlin worker, compiler, target process, or network request.
- **I7 — Privacy:** Local managed evidence may contain bounded compiler identities and repository-relative source paths because agents need them for navigation. It never contains absolute paths, repository URLs, credentials, or unrelated source bodies. Checked-in gate/pilot summaries are stricter: aliases, counts, statuses, and digests only, with no package names, coordinates, paths, or source bodies from private projects.
- **I8 — Independent lifecycle:** Closing or collecting a thread never closes, aborts, publishes, or collects member sessions.
- **I9 — Derived-authority isolation:** S1K-S3K never mutate ThreadAuthority. Every durable result uses binding digest -> evidence CAS -> final authority digest -> deterministic projection, with a direct retained CAS closure.
- **I10 — Profile honesty:** Descriptor facts may prove only their closed compiler projection. They never imply framework routing, runtime behavior, service ownership, HTTP compatibility, or Kotlin source/binary compatibility.

Kotlin Descriptor Navigation v1 contour

Accepted source authority

S1K consumes only sealed `analysis:kotlin-semantic-facts` facts from a `COMPILER_WORKER` generation whose semantic authority is K2, never a syntax-only fallback. Accepted payload schemas are:

- `declaration-descriptor/0.1;`

- `declaration-relation/0.1;`
- `declaration-descriptor-boundary/0.1;`
- `declaration-relation-boundary/0.1.`

The expected authority is `K2_FIR` with `fir-facts-extractor/0.6`. Existing closed semantic validators must be reapplied to every selected payload; the generic `FactRecord` envelope alone is not sufficient authority.

Derived fact model

`codeclew-kotlin-callable-fact/1.0` is a closed tagged union:

- **DECLARATION:** exact member/compilation/generation provenance, repository- relative source anchor and content CAS object, declaration kind, compiler identities, JVM descriptor where applicable, owner/containment, visibility, export boundary, modality, receiver, indexed parameters, return/property type and nullability, type parameters/bounds, and a `shapeDigest` that excludes location and member provenance.
- **USE:** K2-proven source owner, target `CallableId`, supported relation kind, source evidence, and `targetResolution`. Unless a complete exact symbol and same-snapshot dependency authority are present, target resolution is only `CALLABLE_FAMILY`.
- **BOUNDARY:** stable code/stage, member/compilation/subject, evidence reference, and required checks for every K2 or profile-level uncertainty.

Functions and constructors form the exact JVM-callable projection. Classes and properties are supporting declaration shapes; property rows are not described as exact JVM accessor descriptors.

`codeclew-kotlin-callable-fact-set/1.0` binds the thread and thread-context authorities; sorted pair bindings; exact member session/snapshot/generation authorities; compiler, adapter, extractor, runtime, and profile digests; canonical fact shards; counts; budgets; completeness; and a dedicated query- index reference. It is a thread-owned derived object, never a synthetic member generation. To avoid a CAS identity cycle, the query-index manifest binds the pre-publication fact-set `bindingDigest` and exact shard references; the final fact-set authority then binds the resulting query-index CAS object. The index never claims a final fact-set CAS reference that cannot yet exist.

Independent result axes

`shapeStatus` is one of:

- `EXACT_PROJECTED_SHAPE_EQUAL;`
- `EXACT_PROJECTED_SHAPE_DELTA;`
- `UNSURE;`
- `NOT_COMPARABLE.`

`relationshipAuthority` is one of:

- `VERIFIED_SAME_SNAPSHOT_COMPILATION_DEPENDENCY;`
- `DECLARED_TOPOLOGY;`
- `UNBOUND.`

An exact shape never upgrades relationship authority. The frozen real corpus uses `DECLARED_TOPOLOGY` and therefore cannot yield an exact provider/consumer or HTTP claim.

Structural observation IDs

- `KCD_CALLABLE_ADDED`
- `KCD_CALLABLE_REMOVED`
- `KCD_OVERLOAD_SET_CHANGED`
- `KCD_JVM_DESCRIPTOR_CHANGED`

- KCD_PARAMETER_TYPES_CHANGED
- KCD_RETURN_TYPE_CHANGED
- KCD_RECEIVER_TYPE_CHANGED
- KCD_NULLABILITY_CHANGED
- KCD_TYPE_PARAMETER_BOUNDS_CHANGED
- KCD_VISIBILITY_CHANGED
- KCD_MODALITY_CHANGED
- KCD_OVERRIDE_STATUS_CHANGED
- KCD_UNSUPPORTED_COMPARISON

These are compiler-projected observations and planning triggers, never compatibility or breakage verdicts.

Frozen thread-global limits

Resource	Limit
Members	8
Declared pair bindings	32
Selected compilations	64
Input generation facts visited	131,072
Input payload bytes	32 MiB
Normalized declaration facts	65,536
Normalized use facts	65,536
Boundary facts	16,384
Total normalized facts	131,072
Parameters per callable	1,024
Type parameters per declaration	256
Bounds per type parameter	64
Containment depth	64
Identifier/type/path bytes	4,096
Fact shard	8 MiB
Fact/evidence CAS objects per derived authority	64
Retained derived-evidence closure	64 MiB
Query terms	256
Query results	4,096
Impact findings	4,096
Impact obligations	4,096
Source windows	32
Source-window bytes	256 KiB
Coverage document	2 MiB
Coverage-document entries	8,192
Before/after change observations	4,096
Coverage verification obligations	4,096
Coverage validation result rows	8,192
Public compact canonical JSON plus LF	64 KiB

Every atomic construction limit has exact-limit and limit+1 tests. Exceeding one publishes no fact set, index, evidence object, or retained root. Query truncation may return a deterministic prefix only with UNSURE and an explicit narrow/expand obligation.

Canonical dependency graph

S0 — Federated evidence thread

- **Status:** Completed on 2026-08-25; independent review PASS.
- **Outcome:** Bind 2-8 existing heterogeneous sessions under one immutable, path-free ThreadAuthority and produce one globally bounded context with member/session/language/compilation provenance.
- **Implementation:** `thread.rs`, `thread_context.rs`, managed threads/ state, and the original four routes `thread open/context/close/gc`.
- **DoD evidence:** Member-order and semantic-identity tests, exact 8/9 member bounds, evidence/projection substitution rejection, close/publication race, read-only mutation refusal, CAS closure retention, and a mixed Python/Rust warm poison-PATH test all pass. Full Rust acceptance passed 225 tests with one declared release-only test ignored; `clippy -D warnings` passed.

G10 — Historical OpenAPI artifact-fit gate

- **Status:** Completed with `STOP_PROFILE_SELECTION`; independent review PASS.
- **Evidence:** The same ten task IDs and eight pairs were frozen from pinned local topology. Across 11 exact revisions and 338 tracked JSON/YAML candidates, no OpenAPI 3.0/3.1 provider or consumer artifact was found.
- **DoD evidence:** The path-free checked evidence records 0 exact, 0 declared, 10 unusable; its private corpus digest and deterministic selection were independently reproduced. No S1 OpenAPI implementation began.

H1 — Human profile reselection

- **Status:** Completed on 2026-08-25.
- **Decision:** The user selected Kotlin descriptors after being shown that callable/type navigation covered all ten tasks while unresolved service/HTTP relationships would remain UNSURE and Spring semantics would stay deferred.
- **DoD:** This plan preserves G10 as failed historical evidence, states the weaker descriptor outcome explicitly, and introduces G1K before code changes.

G1K — K2 readiness and benchmark-authority gate

- **Status:** Completed with PASS on 2026-08-25.
- **Goal:** Prove the selected profile is executable on every exact pinned unit and freeze a descriptor-oriented value benchmark before S1K implementation.
- **Inputs:** The same ten task IDs, eight pair IDs, topology digest, and 11 exact repository revisions used by G10. G10 classifications are not reused as descriptor classifications.
- **Steps:**
 1. Freeze exact Kotlin compilation selectors for every unit.
 2. Prime each pinned unit through the existing qualified runtime and record sealed generation/query authorities.
 3. Require `COMPILER_WORKER` K2 analysis. Descriptor/relation boundaries may make coverage partial; syntax-only fallback does not qualify.
 4. Prove every task can form a bounded two-member report containing a PROVEN K2 descriptor in an approved file from both members, with at least one callable descriptor and one type descriptor across the task. Relations may supplement this navigation. Descriptor/relation boundaries remain verification obligations and never qualify readiness.

5. Freeze descriptor-specific prompts, per-task callable/file oracle, manual- verification categories, binary rubric, and resource budgets outside Git; publish only their digests and alias/count/status evidence.
6. Require DECLARED_TOPOLOGY, zero endpoint-equivalence scoring, zero HTTP/ Spring exact claims, and no private material in checked evidence.

- **PASS:** All 11 units and all ten two-member tasks satisfy the steps above.

- **FAIL:** Any pinned compilation cannot produce qualified K2 facts, any benchmark authority is missing/mutable, or the rubric scores HTTP equivalence. Failure returns STOP_PROFILE_SELECTION and S1K does not start.

- **Verify:**

```
python3 tools/verify_thread_kotlin_descriptor_gate.py \
docs/plans/evidence/thread-kotlin-descriptor-gate.json
```

- **DoD:** Deterministic corpus/benchmark digests; exact task/pair/revision preservation; all unit/task readiness proved; private files mode 0600; checked evidence path/name/source/credential free; independent audit PASS.

- **DoD evidence:** The canonical v2 run records 11/11 ready units, 10/10 covered tasks, 20/20 qualified task sides, and eight distinct pairs. All qualifying sides use PROVEN K2 descriptors; every task satisfies the callable/type minima; all relationships remain DECLARED_TOPOLOGY; HTTP claims are zero. The independent result audit recomputed the unit, side, task, aggregate, execution, corpus, benchmark, and private-to-checked digests and returned P0=0, P1=0.

S1K — Kotlin callable fact set

- **Status:** Complete; G1K and the S1K implementation are independently verified PASS.

- **Goal:** Normalize existing sealed K2 descriptor/relation facts into one deterministic, bounded, thread-owned callable fact set and dedicated index.

- **Product command:**

```
clew --json thread callables \
--thread THREAD --context THREAD_CONTEXT \
--task-id TASK --pair-id PAIR \
--provider MEMBER --consumer MEMBER --term TERM...
```

- **Implementation outline:**

1. Load only already-ready member generations; never call generation ensure.
2. Bind a closed profile request to the thread/context and sorted declared pairs. Reject unknown members, duplicate/self-substitution, non-Kotlin profile subjects, stale sessions, and mismatched generations.
3. Revalidate every accepted descriptor/relation payload with closed Kotlin semantic validators and exact source CAS/range authority.
4. Emit canonical DECLARATION, USE, and BOUNDARY rows under the frozen limits. Exact lookup uses full symbol identity; token lookup remains navigation-only.
5. Write <=8 MiB canonical shards and a dedicated query index. Bind every direct CAS reference needed after member-session GC.
6. Use the two-phase authority pattern: binding digest -> canonical evidence -> predicted evidence CAS -> final authority -> deterministic projection.
7. Preflight actual compact canonical stdout plus LF, then publish CAS and the thread root while holding thread admission. Partial failure leaves no root.

- **Verify:** focused callable/fact-set/index/thread tests, full Kotlin adapter regression, all-target Rust tests, and clippy -D warnings.

- **DoD:**

- Member/compilation/input-order/shard-order/jobs 1/N permutations produce identical fact-set and index identities.
- Full-symbol lookup distinguishes overloads; lexical collisions and the same CallableId in unrelated repositories never create an exact link.

- Tampered payloads/ranges/JVM descriptors, duplicate identities, stale generations, alias substitution, and profile/runtime mismatch fail closed.
- A partial selected descriptor row/scope, a named boundary relevant to that descriptor, syntax fallback, an ambiguous overload set, a token-only match, or truncation never yields an exact status. Unrelated or subjectless boundaries keep the aggregate result PARTIAL/UNSURE and block absence claims in their affected scope without erasing an independently proved positive descriptor shape.
- Every numeric limit passes at the limit and atomically fails at limit+1.
- Evidence/projection substitution is rejected; a retained root resolves its full direct CAS closure after thread close/GC and member-session GC.
- Warm execution under a poison tool PATH starts no prohibited process.
- Local output is <=64 KiB and contains only required bounded compiler identities and repository-relative source anchors; it contains no absolute path, repository URL, credential, or unrelated source body. Checked-in acceptance evidence additionally omits packages, coordinates, and paths.
- Independent implementation review returns PASS.
- **Acceptance evidence:** Two independent local Kotlin repositories on K2 2.4.10 produced one stable fact-set identity across an initial run and two concurrent repeats under an empty tool PATH. The retained result contains 176 exact projected declarations, 512 non-exact uses, and 392 explicit boundaries/obligations; therefore aggregate coverage correctly remains PARTIAL/UNSURE with DECLARED_TOPOLOGY relationship authority. The three warm callable runs completed in 14.946 s, 14.203 s, and 20.493 s, emitted 2,879 bytes, and started zero prohibited processes. The identical root and reachable-closure digests were independently reconstructed after thread and both member sessions reached terminal GC state. Checked evidence contains no package, coordinate, path, URL, credential, or source-body detail.
- **Reproduce the checked summary:**

```
python3 tools/verify_thread_kotlin_callable_acceptance.py \
docs/plans/evidence/thread-kotlin-callables-acceptance.json
```

S2K — Kotlin descriptor dependency-impact query

- **Status:** Complete; independently verified PASS with no P0/P1 findings.
- **Goal:** Given a bounded subject, return declared provider/consumer members, exact projected callable shapes where support permits, relevant uses/source contexts, and every unresolved verification obligation.
- **Product command:**

```
./clew --json thread impact \
--thread THREAD --fact-set FACT_SET --pair-id PAIR \
--subject-kind full-symbol|callable-family|token --subject SUBJECT \
[--member MEMBER]
```

- **Command decisions:** --thread is required because a fact-set root is retained under one owning thread. The profile is not accepted again because S1K already binds its digest. Subject kind is explicit; --member is required only for a full symbol, whose exact repository namespace cannot be inferred from text. No list/get/expand/status endpoint is added.
- **Implementation outline:**
 1. Consume an immutable S1K fact set; never rebuild it implicitly.
 2. Resolve exact full symbol identities separately from token/CallableId- family navigation.
 3. Report shapeStatus and relationshipAuthority independently. For every frozen cross-repository pair, set relationshipAuthority=DECLARED_TOPOLOGY. Set an exact projected-shape status only when both explicitly selected declarations resolve uniquely and every frozen projected field is supported, complete, and verified; otherwise set shapeStatus=UNSURE. An exact shape status never upgrades the separate relationship authority.
 4. Build an aggregate scope digest from sorted member/profile scopes rather than meeting unrelated member scope digests. Completeness is true only if every required scope is supported, complete, verified, and

untruncated.

5. Fairly select declarations, uses, boundaries, obligations, and source windows. Findings and source windows may truncate only to a deterministic UNSURE prefix. Obligations, retained bytes/CAS count, and the minimum authority projection fail closed; hidden obligations are forbidden.
6. Construct the binding and all CAS references privately, preflight the ≤ 64 KiB projection, then acquire the owning thread admission lock, re-read and re-verify the thread/fact-set authority, publish the full impact evidence, and atomically install its retained root. Partial failure publishes no retained root and a concurrent close wins without resurrection.

• **DoD:**

- Every row resolves to member/session/revision/compilation/generation/fact/ source evidence.
 - Golden exact projected shapes and deltas are found 100%; overloads remain distinct.
 - Coordinate-only, lexical, external-unowned, partial, missing, ambiguous, boundary, or truncated evidence never becomes exact.
 - All ten frozen tasks return both declared members and at least one bounded member finding or explicit subject boundary per side, with zero HTTP/ compatibility claims.
 - Global result/source/stdout budgets fail closed and warm execution starts no prohibited process.
 - Exact-limit and limit+1 tests cover impact findings, obligations, retained bytes, CAS-object count, source windows, and stdout; failure is atomic.
 - Binding/fact-set substitution, partial publication, concurrent close, and GC tests prove that only the admitted impact authority survives and its complete CAS closure remains readable after collection.
 - Independent review returns PASS.
- **Acceptance evidence:** Two independent local Kotlin 2.4.10 repositories produced two identical projected callable declarations. Three relevant compiler boundaries per side correctly lowered the result to `shapeStatus=UNSURE`, while relationship authority remained `DECLARED_TOPOLOGY`. The query returned eight findings, sixteen explicit obligations, and four source windows with no HTTP or compatibility claim. Two direct warm capsule runs with an empty tool path completed in 1.416 s and 1.398 s, emitted byte-identical 13,808-byte output, and started no prohibited process. After the thread and both sessions reached terminal GC state, the retained root and its 567-object closure retained identical digests. The final independent audit ran the full Rust gate (285 library tests passed, one ignored; 11 main and six managed CLI tests passed), Clippy with warnings denied, and diff validation.
- **Reproduce the checked summary:**

```
python3 tools/verify_thread_kotlin_impact_acceptance.py \
docs/plans/evidence/thread-kotlin-impact-acceptance.json
```

S3K — Conditional structural change-set coverage

- **Status:** Complete on 2026-08-25; independently verified PASS with no P0/P1/P2 findings. The original S3K sketch was corrected before implementation by an independent contract audit.
- **Goal:** Compare exact before/after thread/profile authorities and prove whether an inert coverage document acknowledges every observed `KCD_*` change, selected declared pair member, and unresolved obligation.
- **Product command:**

```
./clew --json thread validate \
--before-thread THREAD --before-impact IMPACT \
--after-thread THREAD --after-impact IMPACT \
--member-correspondence BEFORE=AFTER ... \
--coverage FILE
```

- **Corrected v1 contour:** Managed S1K intentionally seals `relationshipAuthority=DECLARED_TOPOLOGY`; therefore S3K v1 has only `VALIDATED_CONDITIONAL` and `INCOMPLETE`. The unreachable `STRUCTURALLY_COVERED` branch is not implemented. It may be introduced only together with a separately qualified upstream same-snapshot dependency authority. S3K accepts only before/after `CALLABLE_FAMILY`

impacts with the same CallableId. Token navigation and namespace-scoped full-symbol impacts cannot form this symmetric comparison.

- **Coverage document:** A closed typed document contains only authority- specific target IDs, their exact verification-category sets, and one inert ACTION or EXTERNAL_WORK tracking ID. Command bodies, notes, source edits, paths, working directories, environment assignments, URLs, and arbitrary objects are not fields in the grammar. Unknown, duplicate, stale, or category-mismatched targets fail before publication; missing targets produce INCOMPLETE. Raw and canonical document bytes are bounded at 2 MiB. The former 1-MiB limit was corrected because the smallest valid closed grammar cannot represent 8,192 entries within it.

- **Implementation outline:**

1. Bind before/after thread, profile, fact-set, impact, member-correspondence, rule, and coverage-document digests. Root the result only under the after thread admission lock; the before thread is immutable input. A comparison digest excludes coverage entries and yields stable target IDs; the final validation binding then adds the canonical document CAS reference, avoiding an identity cycle. Member correspondence is a sorted total bijection over the two members of each selected S2K pair, not over unrelated members of the containing threads. It preserves repository, service, language, compilation, runtime, extractor, adapter, and provider/consumer-role authority. This matches the deliberately pair-scoped S1K/S2K authority and does not claim coverage for thread members outside the comparison.
2. Require an action or explicit external-work marker for every observation, both selected declared pair members, and every conditional obligation.
3. Reject stale/unknown/duplicate anchors, omitted members, shell bodies, source edits, absolute working directories, and executable environment assignments.
4. Return VALIDATED_CONDITIONAL only when every required target and exact verification category is acknowledged; otherwise return INCOMPLETE. Neither status is a compatibility or breakage verdict.
5. Existing plan/task-run/change/publish APIs reject a thread change-set ID.
6. Retain the union of both full callable closures, both selected impact closures, and the two new document/evidence objects. Publish the two new objects in one CAS batch and atomically create the retained root last.

- **DoD:**

- Removing any seeded required member, observation, acknowledgement, or verification category is rejected 100% of the time.
- Every KCD_* rule has before/after golden coverage; none is described as a compatibility/breakage verdict.
- Stale/tampered authorities fail closed; declared-topology evidence can only yield VALIDATED_CONDITIONAL or INCOMPLETE.
- Coverage bytes/entries, observations, obligations, result rows, retained bytes, CAS-object count, and stdout each pass at the exact limit and fail at limit+1. These bounds are exercised at the production pure-core guards; one managed service budget failure proves compositionally that every such prepublication failure installs neither a derived document nor a retained root, without duplicating large synthetic fixtures for each counter.
- Partial publication, before/after/result substitution, concurrent after- thread close, and post-GC retained-closure tests all pass.
- Repeat validation reads CAS only and opens/mutates no repository.
- Full tests, privacy audit, clippy, and independent review pass.
- **Frozen S3K limits:** exactly two member correspondences (2 passes, 3 fails); 2 MiB raw/canonical coverage bytes; 8,192 entries; 4,096 total observations; 4,096 total obligations; 8,192 aggregate member/observation/obligation result rows; exactly two newly derived CAS objects (2 passes, 3 fails); 64 MiB retained closure; and 64 KiB compact public JSON including LF.
- **Acceptance evidence:** A real compiler-backed Kotlin 2.4.10 fixture used two independent local Git repositories and before/after two-member threads. Changing one explicit return from nullable to non-null produced one KCD_NULLABILITY_CHANGED observation while two honest boundary rows remained

KCD_UNSUPPORTED_COMPARISON. Complete inert acknowledgement covered all 39 stable targets and returned VALIDATED_CONDITIONAL, never a compatibility or breakage verdict. Rebuilding the validator capsule changed final authority while preserving the comparison digest, target IDs, and validation binding. Two poisoned-path repeat validations took 4.651 s and 1.755 s, emitted byte-identical 5,052-byte stdout, and started no prohibited process. After both threads and all four before/after sessions reached terminal GC state, the retained root and its 859-object readable closure retained identical digests. The final independent audit passed 34 S3K and 23 S2K focused tests, 320 library tests (one ignored), 13 main CLI tests, seven managed CLI tests, Clippy with warnings denied, and diff validation.

- **Reproduce the checked summary:**

```
python3 tools/verify_thread_kotlin_change_coverage_acceptance.py \
docs/plans/evidence/thread-kotlin-change-coverage-acceptance.json
```

S4K — Frozen pilot and adoption verdict

- **Status:** Not started; depends on S3K.
- **Goal:** Determine whether the complete read-only Kotlin descriptor workflow materially improves agent navigation over equal-budget default local tools.
- **Frozen authority:** The G1K task IDs, pairs, revisions, prompt/oracle/rubric/ budget digests cannot be substituted or amended after S1K begins.
- **Arms:** Same model/configuration/prompt/repositories/revisions; no network; 10-minute wall clock, 40,000 noncached input tokens, and 40 tool starts per task. Both arms may use local Git/rg/file reads. Only Codeclew may use thread/context/callables/impact/expand/validate. Priming is reported but outside the warm measured interval.
- **Metrics:** declared-member coverage, top-10 oracle relevant-file recall, callable/shape correctness, required-manual-check recall, false exact relationship/compatibility claims, elapsed time, source bytes/files opened, tool starts, and noncached agent-input tokens.
- **Binary task pass:** Every oracle-declared member and verification category is named, the primary relevant file is in the top ten, callable/shape claims are correct, and there is no false exact relationship/HTTP/compatibility claim.
- **Warm audit:** Exactly 30 invocations of one primed three-member case; nearest-rank p95 is sorted sample 29.
- **Adoption gate:**
 - fixture exact projected-shape recall 100% and false exact claims zero;
 - frozen-task declared-member recall $\geq 90\%$ and top-10 relevant-file recall $\geq 80\%$;
 - Codeclew passes at least as many tasks as default and omits no critical declared member;
 - median noncached input tokens $\geq 30\%$ lower and median opened files no greater than default;
 - warm p95 ≤ 10 seconds with no Cargo/Rustc/Gradle/Maven/Java/Kotlin worker, cache-copy, target-project process, or network activity;
 - public evidence is path/name/source/credential safe;
 - independent implementation/value review returns PASS.
- **Outcome:** On pass, document a **qualified local Kotlin structural- navigation pilot**. On failure, do not cut over; publish the exact failed metric and the next evidence-based optimization candidate.

Secondary backlog — recorded, not scheduled

Sealed external artifact ownership

Promotion evidence: Repeated tasks require exact cross-repository CallableId ownership and local dependency artifacts can be uniquely bound to provider revisions.

Spring HTTP semantic pack

Promotion evidence: At least three measured misses are specifically caused by missing annotation/composition/configuration meaning, with an approved certainty contour.

OpenAPI/Protobuf/AsyncAPI/JSON Schema

Promotion evidence: A local corpus contains authoritative sealed artifacts covering the target tasks.

FastAPI/Axum/Ktor packs

Promotion evidence: Each language/framework has its own qualified source and certainty authority.

Grouped preparation/coordinated publication

Promotion evidence: Multiple read-only cases prove value and require two qualified mutation sessions plus an approved roll-forward recovery design.

Automatic repository discovery

Promotion evidence: Explicit membership becomes the dominant measured cost and a service-catalog/credential authority is approved.

Fuzzy conceptual search

Promotion evidence: Pilot misses are caused by discovery rather than exact descriptors/declared topology.

Daemon/graph DB/UI/marketplace

Promotion evidence: Warm performance or usability evidence points directly to one of these facilities.

Overall Definition of Done

The revised core plan is complete only when current evidence proves that a user can:

1. open exact local language sessions and bind 2-8 units into one immutable, path-free thread;
2. request one globally bounded cross-repository context;
3. build a deterministic Kotlin callable fact set from already-ready sealed K2 generations without creating a mega-generation;
4. distinguish overloads and inspect exact projected shapes while every external/service relationship remains honestly declared or unsure;
5. query bounded declared-member impact and supporting source evidence;
6. compare before/after projected shapes and catch an omitted member or verification obligation in conditional coverage validation;
7. reproduce every result from retained authorities after close/GC;
8. observe zero thread-driven mutation, target execution, network access, privacy leak, or false HTTP/compatibility claim; and
9. pass the frozen Default-versus-Codeclew adoption gate before documentation recommends this contour.

Independent review reconciliation

One reviewer proposed a common PSI annotation profile. Two architecture/ contract reviews rejected adding that surface to the selected descriptor slice because it would silently introduce Spring/HTTP semantics and still lack consumer artifact ownership. The final reconciled decision is narrower and coherent: reuse existing validated K2 descriptor/relation facts; permit exact projected-shape claims only; retain declared/unsure cross-repository links; and defer annotation/framework semantics. The replacement G1K deliberately does not require exact external artifact ownership because the human selected a conditional navigation profile, but it does require real K2 readiness on every pinned unit and a descriptor-specific frozen benchmark.

Completion checks

```
dot -Tsvg docs/plans/codeclew-multiservice-thread-plan.dot \
-o /tmp/codeclew-multiservice-thread-plan.svg
python3 tools/verify_thread_corpus_gate.py \
docs/plans/evidence/thread-contract-corpus.json
python3 tools/verify_thread_kotlin_descriptor_gate.py \
docs/plans/evidence/thread-kotlin-descriptor-gate.json
python3 tools/verify_thread_kotlin_callables_acceptance.py \
docs/plans/evidence/thread-kotlin-callables-acceptance.json
cargo test -p clew --all-targets --no-fail-fast
cargo clippy -p clew --all-targets -- -D warnings
git diff --check
```


Independent verification record

Status

SUPERSEDED - 2026-08-25.

This file preserves the independent PASS for the pre-H1 OpenAPI version of:

- docs/plans/codeclew-multiservice-thread-plan.md
- docs/plans/codeclew-multiservice-thread-plan.dot

It does not verify the current Kotlin Descriptor Navigation plan selected at H1. The current plan requires its own G1K execution evidence and independent recheck before S1K begins. Its pre-run verdict is recorded separately in multiservice-thread-kotlin-descriptor-plan-verdict.md.

Review chain

1. Two independent architecture reviews examined the current repository and converged on a read-only composition layer above existing single-repository sessions. Their disagreement over the first semantic profile was resolved by choosing one OpenAPI 3.x profile behind a pre-implementation corpus-fit gate.
2. An independent technical verifier first returned CHANGES_REQUIRED with eight concrete authority, evidence, limit, status, command, benchmark, and graph-consistency findings. The plan was revised, rechecked against the same eight findings, and returned PASS.
3. A separate stakeholder-editor reviewed only English clarity, sequence, cohesion, practical rationale, tradeoffs, and approval completeness. Four ambiguities in global budgets, corpus identity, benchmark arms, and inert verification obligations were repaired. The final recheck returned PASS.

Verified properties

- The graph has one dependency order and an explicit stop path.
- Every core step has a practical reason, bounded scope, focused verification, a non-contradiction argument, and an objective Definition of Done.
- Thread, thread-context, contract fact-set, and query-index authorities are separate and immutable.
- Exact contract claims require exact provider and consumer artifacts.
- The OpenAPI surface, structural rule IDs, global limits, and failure behavior are closed before implementation.
- The read-only boundary does not imply candidate execution, behavioral compatibility, coordinated publication, or mutation support.
- The same ten tasks are frozen before implementation and reused for the final Default-versus-Codeclew adoption decision under equal, explicit caps.
- Secondary features are recorded with promotion evidence but are not hidden prerequisites of the core path.

Residual decisions for the human approver

The reviewer still owns the explicit approval choices at the end of the plan: read-only first, composition above sessions, OpenAPI as the single first exact profile subject to G1, local explicit membership, structural-only validation, and the stated adoption thresholds.

There were no unresolved critical or major consistency or readability blockers in the superseded OpenAPI plan. This is historical evidence, not an active approval of the current descriptor profile.